

Jeongjin Roh

professor

基本信息

姓名: Jeongjin Roh

所属院校: 汉阳大学

地区: South Korea

学科领域

集成电路设计与集成系统

微电子科学与工程

电气工程及其自动化

大功率半导体科学与工程

研究领域

模拟与数模混合电路

低功耗电路与能量管理

混合信号集成技术

模拟与射频IC设计

电力电子与电机驱动

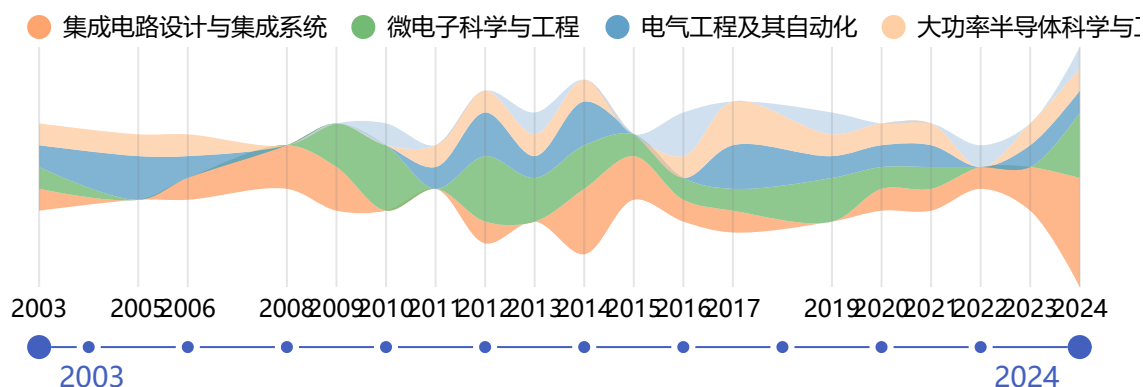
智能功率集成电路

高频电力电子技术

个人简介

Jeongjin Roh is a professor at Hanyang University. His research area is circuit design. He has numerous publications in well - known journals such as IEEE Journal of Solid - State Circuits, IEEE Transactions on Circuits and Systems, etc. Some of his notable works include "A 0.9 - V 60 - 1 - Bit Fourth - Order Delta - Sigma Modulator With 83 - dB Dynamic Range" in 2008 and "A 1.2 - V 4.2 - High - Order Curvature - Compensated CMOS Bandgap Reference" in 2014. He also holds several US patents, like "Chopper stabilized bandgap reference circuit to cancel offset variation" in 2002.

近年发表的论文



A High-PSRR NMOS LDO Regulator With Intrinsic Gain-Tracking Ripple Cancellation Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Jung Sik; Ha, Seunggyun; Jeong, Hongyup; Roh, Jeongjin

被引用次数: 0

2024

A 54-nA Quiescent Current Capless LDO With-39-dB PSRR at 1 MHz Using a Load-Tracking Bandwidth Extension Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Taehyun; Kim, Byeongik; Roh, Jeongjin

被引用次数: 6

2024

High-Precision Built-In Phase Noise Measurement Circuit With a Hybrid $\Delta\Sigma$ Time-to-Digital Converter for SoC Clocking Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Jihun; Na, Sangwook; Kim, Hojin; Roh, Hyungdong; Cho, Youngjae; Choi, Michael; Choo, Min-Seong; Roh, Jeongjin

被引用次数: 0

2024

Automatic Tuning of Negative-R Circuit for High-Performance 95-dB DR 5-kHz Bandwidth Continuous-Time Delta-Sigma Modulator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shin, Hwaseong; Song, Seokjae; Jeong, Hyunji; Duan, Quanzhen; Roh, Jeongjin

被引用次数: 0

2024

A Less-Delay and Wide Conversion-Voltage-Range Level Shifter by Using Switched-Capacitor Technique for Power-Efficiency Health Electronics

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

Duan, Quanzhen; Tao, Tao; Weng, Wuxiong; Huang, Shengming; Liang, Yuhua; Roh, Jeongjin; Xing, Xinpeng

被引用次数: 0

2024

A 464-nA Quiescent Current Buck Converter With 93.7% Peak Efficiency for IoT Devices

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Guo, Tian; Duan, Quanzhen; Lee, Minseok; Roh, Jeongjin

被引用次数: 3

2024

103.1-dB DR Switched-Resistor Delta-Sigma Modulator With Chopped Negative-R for Suppressing Low-Bandwidth Noise

IEEE ACCESS

Shin, Hwaseong; Duan, Quanzhen; Kim, Jaedo; Na, Sangwook; Roh, Jeongjin

被引用次数: 0

2023

A 13.5-nA Quiescent Current LDO With Adaptive Ultra-Low-Power Mode for Low-Power IoT Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Jung Sik; Javed, Khurram; Min, Kyoung Hyun; Roh, Jeongjin

被引用次数: 9

2023

Concurrent Current and Voltage Regulated Buck-Boost Converter for Automotive LED Matrix Headlights

IEEE TRANSACTIONS ON POWER ELECTRONICS

Moon, Jiho; Lee, Jaeseong; Javed, Khurram; Hong, Jupyoo; Roh, Jeongjin

被引用次数: 6

2023

Design of a Low-Power and Area-Efficient LDO Regulator Using a Negative-R-Assisted Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Jung Sik; Javed, Khurram; Roh, Jeongjin

被引用次数: 6

2023

A 2.16- μ W Low-Power Continuous-Time Delta-Sigma Modulator With Improved-Linearity Gm for Wearable ECG Application

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Jaedo; Duan, Quanzhen; Choi, Jihun; Song, Chiho; Roh, Jeongjin

被引用次数: 7

2022

A 0.9- μ A Quiescent Current High PSRR Low Dropout Regulator Using a Capacitive Feed-Forward Ripple Cancellation Technique

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Tian; Kang, Woobin; Roh, Jeongjin

被引用次数: 18

2022

PIFNet: 3D Object Detection Using Joint Image and Point Cloud Features for Autonomous Driving

APPLIED SCIENCES-BASEL

Zheng, Wenqi; Xie, Han; Chen, Yunfan; Roh, Jeongjin; Shin, Hyunchul

被引用次数: 5

2022

A Current-Mode Four-Phase Synchronous Buck Converter With Dynamic Dead-Time Control

IEEE ACCESS

Tang, Jun; Guo, Tian; Kim, Jung Sik; Roh, Jeongjin

被引用次数: 4

2021

100.5 dB SNDR Analog Front-End With a Resistor-Based Discrete-Time Delta-Sigma Modulator for Eliminating Switching Noise and Harmonics

IEEE ACCESS

Song, Seokjae; Kim, Jaedo; Roh, Jeongjin

被引用次数: 7

2021

300 mA LDO Using 0.94 μ A IQ With an Additional Feedback Path for Buffer Turn-off Under Light-Load Conditions

IEEE ACCESS

Jeon, Inho; Guo, Tian; Roh, Jeongjin

被引用次数: 15

2021

A High Slew-Rate Enhancement Class-AB Operational Transconductance Amplifier (OTA) for Switched-Capacitor (SC) Applications

IEEE ACCESS

Kim, Jaedo; Song, Seokjae; Roh, Jeongjin

被引用次数: 17

2020

A 5-MHz bandwidth 78.1-dB SNDR 2-2 MASH delta-sigma modulator

INTERNATIONAL JOURNAL OF ELECTRONICS

Lee, Jaeseong; Song, Seokjae; Roh, Jeongjin

被引用次数: 2

2020

60-V Non-Inverting Four-Mode Buck-Boost Converter With Bootstrap Sharing for Non-Switching Power Transistors

IEEE ACCESS

Moon, Jiho; Lee, Jaeseong; Kim, Seungjin; Ryu, Gyeongha; Hong, Ju-Pyo; Lee, Juhyun; Jin, Haifeng; Roh, Jeongjin

被引用次数: 4

2020

A 20-MHz bandwidth, 75-dB dynamic range, continuous-time delta-sigma modulator with reduced nonidealities

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Song, Seokjae; Lee, Jaeseong; Roh, Jeongjin

被引用次数: 1

2019

Low-Power Fast-Transient Capacitor-Less LDO Regulator With High Slew-Rate Class-AB Amplifier

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tang, Jun; Lee, Jaeseong; Roh, Jeongjin

被引用次数: 57

2019

A 103 dB DR Fourth-Order Delta-Sigma Modulator for Sensor Applications

ELECTRONICS

Lee, Jaeseong; Song, Seokjae; Roh, Jeongjin

被引用次数: 4

2019

A capacitorless low-dropout regulator with enhanced slew rate and 4.5-quiescent current

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Yeo, Jaejin; Javed, Khurram; Lee, Jaeseong; Roh, Jeongjin; Park, Jae-Do

被引用次数: 7

2017

A 17.4-b Delta-Sigma Capacitance-to-Digital Converter for One-Terminal Capacitive Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jung, Youngjae; Duan, Quanzhen; Roh, Jeongjin

被引用次数: 22

2017

Fast-transient DC-DC converter using a high-performance error amplifier with a rapid output-voltage control technique

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Javed, Khurram; Yeo, Jaejin; Lee, Jaeseong; Roh, Jeongjin

被引用次数: 4

2017

A 7.5- μ W 0.08-mm² single-ended SC delta-sigma ADC for acoustic sensor applications

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Duan, Quanzhen; Wang, Zhidong; Roh, Jeongjin

被引用次数: 4

2016

A 1.2V 83dB DR single-ended input SC σ modulator including a large-swing analog buffer for portable ECG applications

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Duan, Quanzhen; Jung, Youngjae; Choi, Danbi; Roh, Jeongjin; Kim, Jongpal

被引用次数: 2

2016

A Delta-Sigma Modulator for Low-Power Analog Front Ends in Biomedical Instrumentation

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Yoon, Younhyun; Duan, Quanzhen; Yeo, Jaejin; Roh, Jeongjin; Kim, Jongjin; Kim, Dongwook

被引用次数: 19

2016

LDO regulator with high power supply rejection at 10MHz

IEICE ELECTRONICS EXPRESS

Javed, Khurram; Roh, Jeongjin

被引用次数: 5

2016

A 1.2-V 4.2-ppm/ $^{\circ}$ C High-Order Curvature-Compensated CMOS Bandgap Reference

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Duan, Quanzhen; Roh, Jeongjin

被引用次数: 122

2015

A 10-MHz multi-bit MASH delta-sigma modulator with analog summing interstage

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Wang, Zhidong; Jung, Youngjae; Roh, Jeongjin

被引用次数: 6

2015

A 0.4 V 63 μ W 76.1 dB SNDR 20 kHz Bandwidth Delta-Sigma Modulator Using a Hybrid Switching Integrator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yoon, Younghyun; Choi, Danbi; Roh, Jeongjin

被引用次数: 23

2015

A 0.03 mm² delta-sigma modulator with cascaded-inverter amplifier

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Wang, Zhidong; Duan, Quanzhen; Roh, Jeongjin

被引用次数: 4

2014

A multiple-output switched-capacitor DC-DC converter for individual brightness control of RGB LEDs with time-interleaved control and output current regulation

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Ahn, Youngkook; Jeon, Inho; Roh, Jeongjin

被引用次数: 0

2014

A True 0.4-V Delta-Sigma Modulator Using a Mixed DDA Integrator Without Clock Boosted Switches

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yoon, Younghyun; Roh, Hyungdong; Roh, Jeongjin

被引用次数: 12

2014

A Multiphase Buck Converter With a Rotating Phase-Shedding Scheme For Efficient Light-Load Control

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ahn, Youngkook; Jeon, Inho; Roh, Jeongjin

被引用次数: 48

2014

A Current Regulator for Inverter-Based Massively Column-Parallel $\Delta\Sigma$ ADCs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yeo, Jaejin; Choi, Yongsuk; Roh, Jeongjin; Han, Gunhee; Chae, Youngcheol; Ham, Seogheon

被引用次数: 11

2014

An Input-Feedforward Multibit Adder-Less Δ - Σ Modulator for Ultrasound Imaging Systems

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Jung, Youngjae; Roh, Hyungdong; Roh, Jeongjin

被引用次数: 17

2013

A single-electrode capacitive sensor using incremental delta-sigma architecture

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Jung, Youngjae; Duan, Quanzhen; Roh, Jeongjin

被引用次数: 1

2013

An 80-V integrated boost converter for piezoelectric actuators in smartphones

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Jeon, Inho; Nam, Hyunseok; Lee, Heeyoung; Ahn, Youngkook; Roh, Jeongjin

被引用次数: 5

2013

A 0.6-V 540-nW delta-sigma modulator for biomedical sensors

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Yoon, Younghyun; Roh, Hyungdong; Lee, Hyuntae; Roh, Jeongjin

被引用次数: 11

2013

A Bufferless Interface for Single-Ended ECM Sensors

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Choi, Youngkil; Roh, Hyungdong; Roh, Jeongjin

被引用次数: 7

2012

5-V Buck Converter Using 3.3-V Standard CMOS Process With Adaptive Power Transistor Driver Increasing Efficiency and Maximum Load Capacity

IEEE TRANSACTIONS ON POWER ELECTRONICS

Nam, Hyunseok; Ahn, Youngkook; Roh, Jeongjin

被引用次数: 51

2012

A 50-MHz Fully Integrated Low-Swing Buck Converter Using Packaging Inductors

IEEE TRANSACTIONS ON POWER ELECTRONICS

Ahn, Youngkook; Nam, Hyunseok; Roh, Jeongjin

被引用次数: 52

2012

A 0.018% THD+N, 88-dB PSRR PWM Class-D Amplifier for Direct Battery Hookup

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Youngkil; Tak, Wonho; Yoon, Younghyun; Roh, Jeongjin; Kwon, Sunwoo; Koh, Jinseok

被引用次数: 40

2012

A buck converter with adaptive on-time PFM control and adjustable output voltage

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Nam, Hyunseok; Ahn, Youngkook; Roh, Jeongjin

被引用次数: 13

2012

A 400-mA current-mode buck converter with a self-trimming current sensing scheme

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Ahn, Youngkook; Heo, Donghun; Nam, Hyunseok; Roh, Jeongjin

被引用次数: 3

2011

Fully synthesised decimation filter for delta-sigma A/D converters

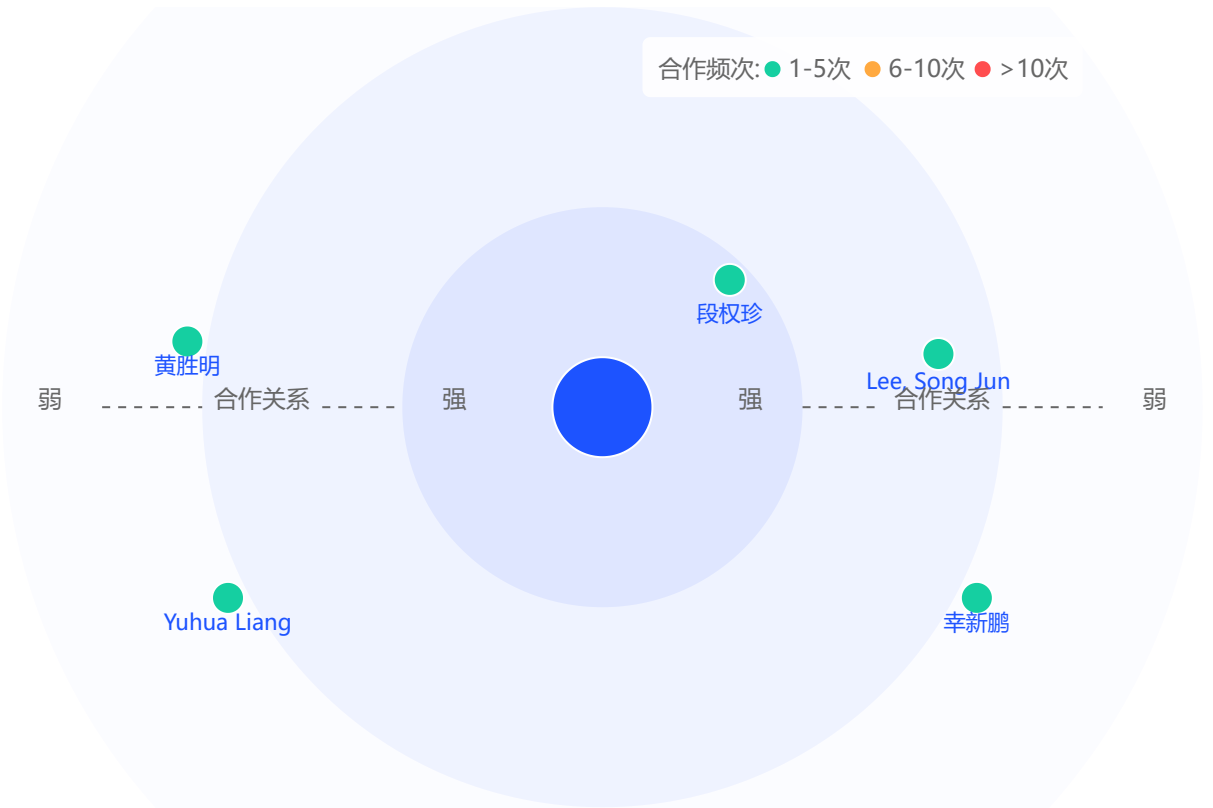
INTERNATIONAL JOURNAL OF ELECTRONICS

Roh, Hyungdong; Byun, Sanho; Choi, Youngkil; Roh, Jeongjin

被引用次数: 4	2010
A 0.8-V 816-nW delta-sigma modulator for low-power biomedical applications ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING Roh, Hyungdong; Lee, Hyuntae; Choi, Youngkil; Roh, Jeongjin 被引用次数: 22	2010
A 89-dB DR 457-μW 20-kHz bandwidth delta-sigma modulator with gain-boosting OTAs ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING Roh, Hyungdong; Choi, Youngkil; Roh, Jeongjin 被引用次数: 3	2010
A 99-dB DR Fourth-Order Delta-Sigma Modulator for 20-kHz Bandwidth Sensor Applications IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT Choi, Youngkil; Roh, Jeongjin; Roh, Hyungdong; Nam, Hyunseok; Lee, Songjun 被引用次数: 21	2009
A 0.6-V Delta-Sigma Modulator With Subthreshold-Leakage Suppression Switches IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS Roh, Hyungdong; Kim, Hyoungjoong; Choi, Youngkil; Roh, Jeongjin; Kim, Yi-Gyeong; Kwon, Jong-Kee 被引用次数: 32	2009
A 0.9-V 60-μW 1-bit fourth-order delta-sigma modulator with 83-dB dynamic range IEEE JOURNAL OF SOLID-STATE CIRCUITS Roh, Jeongjin; Byun, Sanho; Choi, Youngkil; Roh, HyLingdong; Kim, Yi-Gyeong; Kwon, Jong-Kee 被引用次数: 149	2008
Class-D audio amplifier using 1-bit fourth-order delta-sigma modulation IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS Kang, Kyoungsik; Roh, Jeongjin; Choi, Youngkil; Roh, Hyungdong; Nam, Hyunsuk; Lee, Songjun 被引用次数: 30	2008
Digital PWM controller with one-bit noise-shaping interface ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING Roh, Jeongjin 被引用次数: 8	2006
High-gain class-AB OTA with low quiescent current ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING Roh, J 被引用次数: 49	2006
High-performance error amplifier for fast transient DC-DC converters IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS Roh, J	

被引用次数：67	2005
Analogue-to-digital interface technique for digital controllers in DC-DC converters ELECTRONICS LETTERS Roh, J 被引用次数：32005	
A comprehensive signature analysis scheme for oscillation-test IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS Roh, J; Abraham, JA 被引用次数：242003	
Digital PWM controller for DC-DC converters with minimum analogue circuits ELECTRONICS LETTERS Roh, J; Lee, SD; Park, S 被引用次数：52003	

学术合作



段权珍 中山大学	合作次数：4
Lee, Song Jun 汉阳大学	合作次数：2

幸新鹏 中山大学	合作次数：1
Yuhua Liang 西安电子科技大学	合作次数：1
黄胜明 天津理工大学	合作次数：1